Matrices & Matrix Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the standard matrix $[T]$ of the linear operator $T:\mathbb{R}^2\to\mathbb{R}^2$ that performs a horizontal shear by factor 2, followed by a counterclockwise rotation of 45°.

Step 1: Standard matrix of shear $[S]$: $S(x,y)=(x+2y,y)\Rightarrow[S]=(1,2;0,1)$. Step 2: Standard matrix of rotation $[R]$: $[R]=(\cos(45^\circ),-\sin(45^\circ);\sin(45^\circ),\cos(45^\circ))=\frac{\sqrt{2}}{2}(1,-1;1,1)$. Step 3: Compose operators $[T]=[R][S]$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Let $T:\mathbb{R}^2\to\mathbb{R}^2$ have standard matrix $A=(4,1;2,3)$. Find the matrix representation $[T]_{\mathcal{B}}$ relative to the basis $\mathcal{B}=\{(1;1),(-1;2)\}$ via similarity transformation $P^{-1}AP$.

Step 1: Construct change-of-basis transition matrix P: $P=(1,-1;1,2),\quad \det(P)=2-(-1)=3\Rightarrow P^{-1}=\frac{1}{3}(2,1;-1,1)$. Step 2: Compute AP: $AP=(4,1;2,3)(1,-1;1,2)=(4(1)+1(1),4(-1)+1(2);2(1)+3(1),2(-1)+3(2))=(5,-2;5,4)$. Step 3: Compute $[T]_{\mathcal{B}}=P^{-1}(AP)$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Use the Gauss–Jordan elimination algorithm on the augmented block matrix $[A\mid I_3]$ to compute A^{-1} for: $A=(1,0,2;2,-1,3;4,1,8)$.

Step 1: Set up $[A\mid I]$: $(1,0,2,1,0,0;2,-1,3,0,1,0;4,1,8,0,0,1)\to(1,0,2,1,0,0;0,-1,-1,-2,1,0;0,1,0,-4,0,1)$. Step 2: Row swap and scaling: $\to(1,0,2,1,0,0;0,1,0,-4,0,1;0,-1,-1,-2,1,0)\to(1,0,2,1,0,0;0,1,0,-4,0,1;0,0,-1,-6,1,1)$. Step 3: Clear column 3 to reach reduced row echelon form.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Prove that if λ is an eigenvalue of an invertible matrix A, then $\lambda eq 0$ and λ^{-1} is an eigenvalue of A^{-1} with the exact same corresponding eigenvector.

Step 1: Show $\lambda eq 0$: By definition of an eigenvalue, there exists a nonzero vector $\mathbf{v} eq \mathbf{0}$ such that $A\mathbf{v}=\lambda\mathbf{v}$. If $\lambda=0$, then $A\mathbf{v}=\mathbf{0}$, which means $\mathbf{v}\in\text{Null}(A)$, contradicting the assumption that A is invertible. Thus $\lambda eq 0$. Step 2: Show λ^{-1} is an eigenvalue of A^{-1}: Multiply both sides of $A\mathbf{v}=\lambda\mathbf{v}$ by A^{-1} from the left.

5 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Diagonalize the matrix $A=(1,2;2,1)$ by finding invertible P and diagonal D such that $A=PDP^{-1}$. Then compute A^k for any positive integer k.

Step 1: Eigenvalues of A: $\det(A-\lambda I)=(1-\lambda)^2-4=\lambda^2-2\lambda-3=(\lambda-3)(\lambda+1)=0\Rightarrow\lambda_1=3,\ \lambda_2=-1$. Step 2: Eigenvectors:For $\lambda_1=3$: $(-2,2;2,-2)\mathbf{v}_1=\mathbf{0}\Rightarrow\mathbf{v}_1=(1;1)$.For $\lambda_2=-1$: $(2,2;2,2)\mathbf{v}_2=\mathbf{0}\Rightarrow\mathbf{v}_2=(-1;1)$. Step 3: Construct P, D, and P^{-1}.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the matrix $A = (1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5)$, find bases for $\text{Row}(A)$, $\text{Col}(A)$, and $\text{Null}(A)$. Verify the Rank–Nullity Theorem.

Step 1: Row reduction to RREF: $(1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5) \to (1, 2, 0, 1; 0, 0, 1, 2; 0, 0, 1, 2) \to (1, 2, 0, 1; 0, 0, 1, 2; 0, 0, 0, 0)$. Step 2: Subspace Bases:Basis for $\text{Row}(A)$: Non-zero rows of RREF: $\{(1, 2, 0, 1), (0, 0, 1, 2)\}$.Basis for $\text{Col}(A)$: Pivot columns of original matrix A (columns 1 and 3): $\{(1; 2; 3), (0; 1; 1)\}$.Basis for $\text{Null}(A)$: Free variables.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = (1; 2)$. If the trace of A is 7, find the other eigenvalue of A.

Let the matrix A be $(a, b; c, d)$. The eigenvalues of A are the roots of the characteristic equation $\det(A - \lambda I) = 0$. We are given that $\lambda_1 = 3$ is an eigenvalue. The trace of a matrix is the sum of its diagonal elements, so $\text{tr}(A) = a + d$. We are given that $\text{tr}(A) = 7$. The trace of a matrix is also equal to the sum of its eigenvalues. Let the eigenvalues be λ_1 and λ_2. Therefore, $\lambda_1 + \lambda_2 = \text{tr}(A)$. Substituting the given.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $A = (1, 2, 0; 0, 3, 1; 2, 1, 4)$. Compute $\det(A)$, find the adjugate matrix $\text{adj}(A)$, and verify the identity $A\text{adj}(A) = \det(A)I_3$.

Step 1: Compute $\det(A)$: Expanding along row 1: $\det(A) = 1(3(4) - 1(1)) - 2(0(4) - 1(2)) + 0 = 1(11) - 2(-2) = 11 + 4 = 15$. Step 2: Compute Cofactor Matrix C: $C_{11} = +(12 - 1) = 11, \quad C_{12} = -(0 - 2) = 2, \quad C_{13} = +(0 - 6) = -6$ $C_{21} = -(8 - 0) = -8, \quad C_{22} = +(4 - 0) = 4, \quad C_{23} = -(1 - 4) = 3$ $C_{31} = +(2 - 0) = 2, \quad C_{32} = -(1 - 0) = -1, \quad C_{33} = +(3 - 0) = 3$ The adjugate matrix is the transpose $\text{adj}(A) = C^T$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let A and B be invertible $n \times n$ matrices. Prove algebraically that: (a) $(AB)^T = B^T A^T$, and (b) $(AB)^{-1} = B^{-1}A^{-1}$.

(a) Transpose Proof: The (i, j)-entry of $(AB)^T$ is the (j, i)-entry of AB: $[(AB)^T]_{ij} = [AB]_{ji} = \sum_{n}^{n} A_{jk}B_{ki} = \sum_{n}^{n} B_{ki}A_{jk}$. Using the definition of matrix transpose, $B_{ki} = [B^T]_{ik}$ and $A_{jk} = [A^T]_{kj}$: $[(AB)^T]_{ij} = \sum_{k=1}^{n} [B^T]_{ik}[A^T]_{kj} = [B^T A^T]_{ij}$. Since entries match for all i, j, $(AB)^T = B^T A^T$. ■ (b) Inverse Proof: A matrix C is the inverse of AB if and only if $(AB)C = I_n$ and $C(AB) = I_n$. Test.

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10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use Cramer's Rule to solve for x_2 in the linear system: $\{2x_1 + x_2 + x_3 = 4; x_1 - x_2 + 2x_3 = 1; 3x_1 + 2x_2 - x_3 = 7$

Step 1: Compute main determinant $\det(A)$: $\det(A) = |2, 1, 1; 1, -1, 2; 3, 2, -1| = 2((-1)(-1) - 2(2)) - 1(1(-1) - 2(3)) + 1(1(2) - (-1)(3))$ $= 2(1 - 4) - 1(-1 - 6) + 1(2 + 3) = 2(-3) - 1(-7) + 1(5) = -6 + 7 + 5 = 6.$ Step 2: Replace column 2 with $\mathbf{b} = (4, 1, 7)^T$: $\det(A_2(\mathbf{b})) = |2, 4, 1; 1, 1, 2; 3, 7, -1|.$ Expanding along row 1: $= 2(1(-1) - 2(7)) - 4(1(-1) - 2(3)) + 1(1(7) - 1(3)) = 2(-15) - 4(-7) + 1(4) = -30 + 28 + 4 = 2.$ Step 3: Evaluate x_2.

11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Compute the determinant of the 4×4 matrix by Laplace cofactor expansion along an optimal row or column: $A = (2, 0, -1, 3; 0, 5, 0, 0; 1, -2, 3, 1; 4, 0, 1, 2)$.

Step 1: Choose row with maximum zeros: Row 2 has three zero entries: $a_{21} = 0, a_{22} = 5, a_{23} = 0, a_{24} = 0$. Expanding along Row 2: $\det(A) = (-1)^{2+2}(5)|2, -1, 3; 1, 3, 1; 4, 1, 2| = 5|2, -1, 3; 1, 3, 1; 4, 1, 2|.$ Step 2: Evaluate 3×3 determinant: $|2, -1, 3; 1, 3, 1; 4, 1, 2| = 2(3(2) - 1(1)) - (-1)(1(2) - 1(4)) + 3(1(1) - 3(4))$ $= 2(6 - 1) + 1(2 - 4) + 3(1 - 12) = 2(5) + 1(-2) + 3(-11) = 10 - 2 - 33 = -25.$ Step 3: Multiply by coefficient.

12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the matrix $A = (2, 1; 1, 3)$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $(x'; y') = A(x; y)$, what is the image of the point $(3, 1)$ under this transformation?

The transformation is defined by the matrix multiplication $(x'; y') = A(x; y)$. We are given the point $(3, 1)$, so $(x; y) = (3; 1)$ and the matrix $A = (2, 1; 1, 3)$. Performing the matrix multiplication: $(x'; y') = (2, 1; 1, 3)(3; 1) = ((2 \times 3) + (1 \times 1); (1 \times 3) + (3 \times 1)) = (6 + 1; 3 + 3) = (7; 6)$ Therefore, the image of the point $(3, 1)$ is $(7, 6)$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the component orthogonal to $\mathbf{u}$ of $\mathbf{y} = \langle -2, 3, -4 \rangle$ for $\mathbf{u} = \langle 1, -3, 3 \rangle$.

$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \frac{-23}{19} \mathbf{u} = \langle -\frac{23}{19}, \frac{69}{19}, -\frac{69}{19} \rangle$, so $\mathbf{y} - \hat{\mathbf{y}} = \langle -\frac{15}{19}, -\frac{12}{19}, -\frac{7}{19} \rangle$. Final Answer: $\langle -\frac{15}{19}, -\frac{12}{19}, -\frac{7}{19} \rangle$

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Prove that if $\mathbf{x}_p$ is any particular solution to $A\mathbf{x} = \mathbf{b}$, then the complete solution set is the affine subspace $S = \{\mathbf{x}_p + \mathbf{x}_h : \mathbf{x}_h \in \text{Null}(A)\}$.

Two-part set inclusion proof: 1. Show $S \subseteq \text{Solution Set}$: Let $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$ where $A\mathbf{x}_p = \mathbf{b}$ and $A\mathbf{x}_h = \mathbf{0}$. Applying matrix multiplication: $A\mathbf{x} = A(\mathbf{x}_p + \mathbf{x}_h) = A\mathbf{x}_p + A\mathbf{x}_h = \mathbf{b} + \mathbf{0} = \mathbf{b}$. Thus every vector in S is indeed a valid solution. 2. Show $\text{Solution Set} \subseteq S$: Let $\mathbf{y}$ be an arbitrary solution to $A\mathbf{x} = \mathbf{b}$, so.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Find the shortest Euclidean distance from the point $P(3, -1, 4)$ to the hyperplane $\mathcal{H}: 2x - y + 2z = 5$ in $\mathbb{R}^3$.

Hyperplane Distance Formula: For $\mathcal{H}: \mathbf{n} \cdot \mathbf{x} = d$, the perpendicular distance from $\mathbf{x}_0$ is: $D = \frac{|\mathbf{n} \cdot \mathbf{x}_0 - d|}{||\mathbf{n}||}$. Here $\mathbf{n} = (2, -1, 2)^T$, $d = 5$, and $\mathbf{x}_0 = (3, -1, 4)^T$. 1. Norm of normal vector: $||\mathbf{n}|| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$. 2. Value at point P: $\mathbf{n} \cdot \mathbf{x}_0 = 2(3) - 1(-1) + 2(4) = 6 + 1 + 8 = 15$. 3. Distance.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Determinants

Find all eigenvalues, their algebraic and geometric multiplicities, and a basis for each eigenspace for: $A = (3, 1, 0; 0, 3, 0; 0, 0, 2)$. Is A diagonalizable?

Step 1: Characteristic equation: Since A is upper triangular, its eigenvalues are the diagonal entries: $\det(A - \lambda I) = (3 - \lambda)^2(2 - \lambda) = 0$. Distinct eigenvalues: $\lambda_1 = 3$ (algebraic multiplicity $\alpha_1 = 2$), and $\lambda_2 = 2$ (algebraic multiplicity $\alpha_2 = 1$). Step 2: Eigenspace for $\lambda = 3$: $(A - 3I)\mathbf{v} = (0, 1, 0; 0, 0, 0; 0, 0, -1)(v_1; v_2; v_3) = (0; 0; 0)$. We obtain $v_2 = 0$ and $-v_3 = 0 \Rightarrow v_3 = 0$, w.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Vectors in n Dimensions

Apply the Gram–Schmidt orthogonalization process to the columns of A to find its QR factorization: $A = (1, 2; 2, 1; -2, 2)$.

Step 1: Orthogonalize $\mathbf{x}_1 = (1, 2, -2)^T$ and $\mathbf{x}_2 = (2, 1, 2)^T$: First vector: $\mathbf{v}_1 = \mathbf{x}_1 = (1; 2; -2)$. Norm: $||\mathbf{v}_1|| = \sqrt{1^2 + 2^2 + (-2)^2} = \sqrt{9} = 3$. Subtract projection of $\mathbf{x}_2$ onto $\mathbf{v}_1$: $\mathbf{x}_2 \cdot \mathbf{v}_1 = 2(1) + 1(2) + 2(-2) = 2 + 2 - 4 = 0$. Since $\mathbf{x}_2 \cdot \mathbf{v}_1 = 0$, the columns were already orthogonal! Thus $\mathbf{v}_2 = \mathbf{x}_2 = (2; 1; 2)$. Norm.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Perform a complete bifurcation analysis of the system in terms of parameter $k \in \mathbb{R}$: $\{x + y + kz = 2; 3x + 4y + 2z = k; 2x + 3y - z = 1$ Determine all values of k for which the system has (a) a unique solution, (b) infinitely many solutions, and (c) no solution.

Step 1: Compute determinant of coefficient matrix A: $\det(A) = |1, 1, k; 3, 4, 2; 2, 3, -1|$. Expanding along row 1: $\det(A) = 1(4(-1) - 2(3)) - 1(3(-1) - 2(2)) + k(3(3) - 4(2)) = 1(-10) - 1(-7) + k(1) = k - 3$. Step 2: Classify cases:(a) Unique Solution: If $keq3$, $\det(A)eq0$; the system has full rank 3 and possesses a unique solution.Critical Case $k = 3$: Substitute $k = 3$ into the augmented matrix.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Determinants

Level 3 · Advanced / University RigorSuppose an $n \times n$ matrix A satisfies the polynomial equation $A^3 - 4A^2 + 2A - 5I = 0$. Prove that A is invertible, and express A^{-1} as an explicit polynomial in A.

Step 1 · Isolate the Identity Matrix: $A^3 - 4A^2 + 2A = 5I$. Divide both sides by 5: $\frac{1}{5}(A^3 - 4A^2 + 2A) = I$.
Step 2 · Factor out A: $A\left[\frac{1}{5}(A^2 - 4A + 2I)\right] = I$. Similarly, since polynomials in A commute: $\left[\frac{1}{5}(A^2 - 4A + 2I)\right]A = I$. Conclusion: By the definition of matrix inverse, A is invertible, and its inverse is: $\mathbf{A^{-1} = \frac{1}{5}A^2 - \frac{4}{5}A + \frac{2}{5}I}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Vectors in n Dimensions

Level 2 · MediumLet $A = (3, 5; 1, 2)$ and $\mathbf{b} = (4; 1)$. Compute A^{-1} and use it to solve the linear system $A\mathbf{x} = \mathbf{b}$.

Step 1 · Invert A: $\det(A) = (3)(2) - (5)(1) = 6 - 5 = 1$. $A^{-1} = \frac{1}{1}(2, -5; -1, 3) = (2, -5; -1, 3)$.
Step 2 · Multiply $\mathbf{x} = A^{-1}\mathbf{b}$: $\mathbf{x} = (2, -5; -1, 3)(4; 1) = (2(4) - 5(1); -1(4) + 3(1)) = (\mathbf{3; -1})$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If matrix A has size 3×4 and matrix B has size 4×2, what is the size of the product AB?

a. 3×2 ✓

b. 4×4

c. 2×3

d. Multiplication is undefined

An $m \times k$ matrix multiplied by a $k \times n$ matrix produces an $m \times n$ matrix. Here $(3 \times 4) \times (4 \times 2) = 3 \times 2$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Is matrix multiplication commutative in general ($AB = BA$)?

a. **No, $AB \neq BA$ in general even when both products are defined** ✓

b. Yes, multiplication of numbers and matrices is always commutative

c. Only if the entries are integers

d. Yes, provided both matrices are square

Matrix multiplication is fundamentally non-commutative: in general, $AB \neq BA$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
What is $(AB)^T$ equal to in terms of the transposes A^T and B^T?

a. B^TA^T ✓

b. A^TB^T

c. AB^T

d. BA^T

The transpose of a product reverses order: $(AB)^T = B^TA^T$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Find the inverse of the 2×2 matrix $A = [[4, 7], [1, 2]]$.

a. $[[2, -7], [-1, 4]]$ ✓

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

$\det(A) = 4(2) - 7(1) = 8 - 7 = 1$. The inverse is $\frac{1}{1}[[2, -7], [-1, 4]] = [[2, -7], [-1, 4]]$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
What is an elementary matrix?

a. **A matrix obtained by performing a single elementary row operation on an identity matrix** ✓

b. Any matrix containing only 0s and 1s

c. A diagonal matrix with determinant 1

d. A matrix with exactly one non-zero row

An elementary matrix E represents a single row operation; multiplying EA performs that row operation on A.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
If A, B, C are invertible square matrices, what is $(ABC)^{-1}$?

a. $C^{-1}B^{-1}A^{-1}$ ✓

b. $A^{-1}B^{-1}C^{-1}$

c. CBA

d. ABC^{-1}

Inverting a product reverses the sequence of all factors: $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A square matrix A is symmetric if and only if:

a. $A^T = A$ ✓

b. $A^T = -A$

c. $A^T = A^{-1}$

d. $\det(A) = 1$

A matrix is symmetric when it equals its own transpose: $A^T = A$ ($a_{ij} = a_{ji}$ for all i, j).

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What defines an orthogonal matrix Q?

a. $Q^TQ = I$, **meaning** $Q^{-1} = Q^T$ ✓

b. $Q^2 = I$

c. $\det(Q) = 0$

d. All entries of Q are perpendicular

An orthogonal matrix has orthonormal columns, which means $Q^TQ = I$ and therefore $Q^{-1} = Q^T$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If Q is an orthogonal matrix, what are the possible values for $\det(Q)$?

a. ± 1 ✓

b. 0

c. 1 only

d. Any real number

$\det(Q^TQ) = \det(I) = 1 \Rightarrow (\det Q)^2 = 1 \Rightarrow \det(Q) = \pm 1$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A transformation $T: \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if for all vectors $\mathbf{u}$, $\mathbf{v}$ and scalars c:

a. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ **and** $T(c\mathbf{u}) = cT(\mathbf{u})$ ✓

b. $T(\mathbf{u} \cdot \mathbf{v}) = T(\mathbf{u}) \cdot T(\mathbf{v})$

c. $T(\mathbf{u}) = ||\mathbf{u}||$

d. $T(\mathbf{0}) = [1, 1, \ldots, 1]^T$

Linearity is defined by preservation of vector addition and scalar multiplication: $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the standard matrix for counterclockwise rotation in $\mathbb{R}^2$ by angle θ?

a. $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$ ✓

b. $[[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$

c. $[[\sin\theta, \cos\theta], [\cos\theta, -\sin\theta]]$

d. $[[\cos\theta, 0], [0, \sin\theta]]$

Rotating $[1, 0]^T$ gives $[\cos\theta, \sin\theta]^T$, and rotating $[0, 1]^T$ gives $[-\sin\theta, \cos\theta]^T$, giving $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the standard matrix for reflection across the x-axis in $\mathbb{R}^2$?

a. $[[1, 0], [0, -1]]$ ✓

b. $[[-1, 0], [0, 1]]$

c. $[[0, 1], [1, 0]]$

d. $[[-1, 0], [0, -1]]$

Reflecting across the x-axis keeps x the same and negates y: $T([x, y]^T) = [x, -y]^T$, so the matrix is $[[1, 0], [0, -1]]$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the matrix $A = (2, 1; 3, 4)$. What is the matrix A^T?

a. $(2, 3; 1, 4)$ ✓

b. $(1, 2; 4, 3)$

c. $(4, 1; 3, 2)$

d. $(2, 4; 1, 3)$

The transpose of a matrix A, denoted by A^T, is obtained by interchanging its rows and columns. For a matrix $A = (a, b; c, d)$, its transpose is $A^T = (a, c; b, d)$. Given the matrix $A = (2, 1; 3, 4)$, interchanging the rows and columns yields $A^T = (2, 3; 1, 4)$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the matrix $A = (2, 0; 0, 3)$. If $\mathbf{v} = (1; 2)$, what is $A\mathbf{v}$?

a. $(2; 3)$

b. $(1; 6)$

c. $(2; 6)$ ✓

d. $(3; 2)$

To find $A\mathbf{v}$, we perform matrix multiplication. \\ $A\mathbf{v} = (2, 0; 0, 3)(1; 2) = ((2)(1) + (0)(2); (0)(1) + (3)(2)) = (2 + 0; 0 + 6) = (2; 6)$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Which matrix is orthogonal (satisfies $Q^TQ = I$)?

a. $[(1, 1; 0, 1)]$

b. $[(\frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2})]$

c. $[(0, -1; 1, 0)]$ ✓

d. $[(1, -1; 1, 1)]$

The columns of $[(0, -1; 1, 0)]$ are orthonormal, so $Q^TQ = I$. Each other matrix has columns that are not orthogonal or not unit length.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A 3×3 matrix has eigenvalues $-5, 2, 2$. Find its determinant.

a. -20 ✓

b. -1

c. -19

d. 20

The determinant is the product of the eigenvalues (with multiplicity): -20.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A 2×2 matrix has eigenvalues $-1, -5$. Find its trace.

a. -6 ✓

b. 5

c. -5

d. 6

The trace is the sum of the eigenvalues (with multiplicity): -6.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Systems of Linear Equations

A parallelogram in the xy-plane is defined by the vertices $(0, 0)$, $(3, 1)$, $(1, 4)$, and $(4, 5)$. What is the area of this parallelogram?

a. 11

b. **13** ✓

c. 15

d. 17

The area of a parallelogram formed by vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ originating from the same point is given by the absolute value of the determinant of the matrix formed by these vectors: $|\det([u_1, v_1\ u_2, v_2])|$. In this case, we can take the vectors originating from $(0, 0)$ to be $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (1, 4)$. The area is $|\det([3, 1\ 1, 4])| = |(3 \times 4) - (1 \times 1)| = |12 - 1| = |11| = 11$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Determinants

A linear map T satisfies $T\langle -1, -1\rangle = \langle 0, -2\rangle$ and $T\langle 0, -1\rangle = \langle 2, -1\rangle$. Find $T\langle 2, -3\rangle$.

a. $\langle -6, -1\rangle$

b. $\langle -10, 1\rangle$

c. $\langle 10, -1\rangle$ ✓

d. $\langle 11, -1\rangle$

Write $\langle 2, -3\rangle = c_1\langle -1, -1\rangle + c_2\langle 0, -1\rangle$ with $(c_1, c_2) = \langle -2, 5\rangle$. By linearity, $T\langle 2, -3\rangle = c_1\langle 0, -2\rangle + c_2\langle 2, -1\rangle = \langle 10, -1\rangle$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the image of $\langle 1, 1\rangle$ under the rotation counterclockwise by 30°.

a. $\langle -\frac{1}{2} + \frac{\sqrt{3}}{2}, \frac{1}{2} + \frac{\sqrt{3}}{2}\rangle$ ✓

b. $\langle \frac{1}{2} + \frac{\sqrt{3}}{2}, -\frac{1}{2} + \frac{\sqrt{3}}{2}\rangle$

c. $\langle \frac{1}{2} - \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2} - \frac{1}{2}\rangle$

d. $\langle 1, 1\rangle$

The standard matrix is $\left[\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}; \frac{1}{2}, \frac{\sqrt{3}}{2}\right)\right]$; multiplying gives $\langle -\frac{1}{2} + \frac{\sqrt{3}}{2}, \frac{1}{2} + \frac{\sqrt{3}}{2}\rangle$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Systems of Linear Equations

Which set is NOT orthonormal?

a. $\{\langle \frac{5}{13}, \frac{12}{13}\rangle, \langle \frac{12}{13}, -\frac{5}{13}\rangle\}$

b. $\{\langle \frac{1}{2}, \frac{1}{2}\rangle, \langle \frac{1}{2}, -\frac{1}{2}\rangle\}$ ✓

c. $\{\langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3}\rangle, \langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\rangle\}$

d. $\{\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\rangle, \langle \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\rangle\}$

$\{\langle \frac{1}{2}, \frac{1}{2}\rangle, \langle \frac{1}{2}, -\frac{1}{2}\rangle\}$ fails: either the dot product is nonzero or a vector does not have length 1. The other sets pass both tests.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Determinants

Apply Gram–Schmidt to $\mathbf{x}_1 = \langle 1, -1\rangle$, $\mathbf{x}_2 = \langle 2, 3\rangle$. Which vector is parallel to the resulting $\mathbf{v}_2$?

a. $\langle 2, 3\rangle$

b. $\langle \frac{3}{2}, \frac{7}{2}\rangle$

c. $\langle 1, 1\rangle$ ✓

d. $\langle 1, -1\rangle$

$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}\mathbf{x}_1 = \langle \frac{5}{2}, \frac{5}{2}\rangle$, a multiple of $\langle 1, 1\rangle$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the least-squares line for the points $(-1,\ 3), (1,\ -3), (3,\ 1), (4,\ 1)$.

a. $y = \frac{47}{59} - \frac{10x}{59}$ ✓

b. $y = \frac{47x}{59} - \frac{10}{59}$

c. $y = \frac{106}{59} - \frac{10x}{59}$

d. $y = \frac{10x}{59} + \frac{47}{59}$

Solve the normal equations $A^TA = A^Ty$ with $A^TA = [(4, 7; 7, 27)]$, $A^T\mathbf{y} = \langle 2, 1\rangle$: $\hat{\beta}_0 = \frac{47}{59},\ \hat{\beta}_1 = -\frac{10}{59}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Systems of Linear Equations

Which vector spans the kernel of $T(\mathbf{x}) = [(-2, 3, 0; 3, -1, 1)]\mathbf{x}$?

a. $\langle -2, -3, 7\rangle$

b. $\langle -3, 2, 7\rangle$

c. $\langle -3, -2, 7\rangle$ ✓

d. $\langle 1, 1, 1\rangle$

Row-reduce and solve $A\mathbf{x} = \mathbf{0}$: one free variable, and the solutions are the multiples of $\langle -3, -2, 7\rangle$. (Check: $A\langle -3, -2, 7\rangle = \mathbf{0}$.)

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Determinants

Find the y-intercept of the least-squares line for $(-2,\ -2), (1,\ -3), (3,\ 6), (4,\ -3)$.

a. $\frac{10}{21}$

b. $-\frac{17}{14}$ ✓

c. $-\frac{3}{14}$

d. $\frac{17}{14}$

Normal equations: $[(4, 6; 6, 30)][\beta_0; \beta_1] = [-2; 7]$, giving $\hat{\beta}_0 = -\frac{17}{14},\ \hat{\beta}_1 = \frac{10}{21}$.

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